
Robotics & State Estimation (RiSE) Tutorials

Lecture Notes

Yulin Yang - yangyulin1@gmail.com
Chuchu Chen - chuchu.chen@gwu.edu
Xingxing Zuo - xingxing.zuo@mbzuai.ac.ae

Robotics & State Estimation Tutorials

<https://github.com/yangyulin/rise-tutorial>



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1 Lecture 1: Foundations — Probability, Linear Algebra, and Linear Systems

1.1 Probability

1.1.1 Definition

For an event A , the probability $P(A)$ is

$$P(A) = \frac{N_A}{N}, \quad (1)$$

where N_A is the number of occurrences of A and N is the number of all possible outcomes.

1.1.2 Conditional probability

The probability of event A given that event B is true is

$$P(A | B) = \frac{P(A, B)}{P(B)}, \quad P(B) \neq 0, \quad (2)$$

which yields the product rule

$$P(A, B) = P(A | B) P(B) \quad (3)$$

$$= P(B | A) P(A). \quad (4)$$

1.1.3 Total probability theorem

If $\{A_1, \dots, A_n\}$ is a partition (mutually exclusive, $P(A_i, A_j) = 0$ for $i \neq j$) and B is an arbitrary event, then

$$P(B) = P(B | A_1)P(A_1) + \dots + P(B | A_n)P(A_n) = \sum_i P(B | A_i) P(A_i). \quad (5)$$

1.1.4 Bayes' theorem

$$P(A_i | B) = \frac{P(B | A_i) P(A_i)}{P(B)} = \frac{P(B | A_i) P(A_i)}{\sum_i P(B | A_i) P(A_i)}. \quad (6)$$

1.1.5 Independence

With the inclusion–exclusion relation $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, two events A and B are independent when

$$P(A, B) = P(A | B) P(B) \quad (7)$$

$$= P(B | A) P(A) \quad (8)$$

$$= P(A) P(B) = P(B) P(A). \quad (9)$$

1.2 Random Variables

A random variable defines a function that quantifies an experiment. For a sample space $S = \{s_1, \dots, s_n\}$, a random variable is a map $x : S \rightarrow \mathbb{R}$ with $x(s_i) = x_i$.

1.2.1 Distribution

The (cumulative) distribution of a random variable x is

$$F_x(x) = P(x \leq x), \quad (10)$$

i.e. the probability that x takes a value lower than or equal to x .

1.2.2 Probability density function (pdf)

$$p(x) = \lim_{\varepsilon \rightarrow 0} \frac{F_x(x) - F_x(x - \varepsilon)}{\varepsilon} \quad (11)$$

$$= \frac{dF_x(x)}{dx} = \frac{dP(x \leq x)}{dx}, \quad (12)$$

hence

$$P(x \leq x) = \int_{-\infty}^x p(\xi) d\xi, \quad P(x \leq \infty) = 1 = \int_{-\infty}^{+\infty} p(\xi) d\xi, \quad (13)$$

and for an interval

$$P(x_1 \leq x < x_2) = \int_{x_1}^{x_2} p(\xi) d\xi. \quad (14)$$

1.2.3 Example distributions

- Gaussian: $x \sim \mathcal{N}(m, \sigma) \iff p(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-m)^2}{2\sigma^2}}.$

- Uniform: $p(x) = \begin{cases} 0 & x < x_1 \\ \frac{1}{x_2 - x_1} & x_1 \leq x \leq x_2 \\ 0 & x > x_2 \end{cases}$

1.2.4 Conditional distribution and density

$$P(x \leq x | A) = \frac{P(x \leq x, A)}{P(A)}, \quad p(x | A) = \frac{dP(x \leq x | A)}{dx}. \quad (15)$$

1.2.5 Total probability theorem for pdfs

For a mutually exclusive partition $\{A_1, \dots, A_n\}$,

$$p(x) = p(x | A_1)P(A_1) + \dots + p(x | A_n)P(A_n). \quad (16)$$

1.2.6 Continuous Bayes' theorem

Using the infinitesimal relations $P(\mathbf{x} = x) = p(x) \delta x$ and $P(\mathbf{x} = x | A) = p(x | A) \delta x$,

$$P(A | \mathbf{x} = x) = \frac{P(A, \mathbf{x} = x)}{P(\mathbf{x} = x)} \quad (17)$$

$$= \frac{P(\mathbf{x} = x | A) P(A)}{P(\mathbf{x} = x)} \quad (18)$$

$$= \frac{p(x | A)}{p(x)} P(A), \quad (19)$$

so that $P(A | \mathbf{x} = x) p(x) = p(x | A) P(A)$, which gives the continuous total-probability and Bayes' relations

$$P(A) = \int_{-\infty}^{+\infty} P(A | \mathbf{x} = x) p(x) dx, \quad (20)$$

$$p(x | A) = \frac{P(A | \mathbf{x} = x)}{P(A)} p(x) \quad (21)$$

$$= \frac{P(A | \mathbf{x} = x) p(x)}{\int_{-\infty}^{+\infty} P(A | \mathbf{x} = x) p(x) dx}. \quad (22)$$

1.3 Expectation, Variance, and Random Vectors

1.3.1 Mean

The expected value (mean) of a continuous random variable, and its conditional version, are

$$m_x = \mathbb{E}(\mathbf{x}) = \int_{-\infty}^{+\infty} x p(x) dx, \quad m_x = \mathbb{E}(\mathbf{x} | A) = \int_{-\infty}^{+\infty} x p(x | A) dx. \quad (23)$$

1.3.2 Variance

$$\sigma_x^2 = \mathbb{E}((\mathbf{x} - m_x)^2) \quad (24)$$

$$= \int_{-\infty}^{+\infty} (x - m_x)^2 p(x) dx \quad (25)$$

$$= \mathbb{E}(\mathbf{x}^2) - (\mathbb{E}(\mathbf{x}))^2, \quad (26)$$

with the conditional variance

$$\sigma_{x|A}^2 = \mathbb{E}\{(\mathbf{x} - m_{x|A})^2 | A\} = \int_{-\infty}^{+\infty} (x - m_{x|A})^2 p(x | A) dx. \quad (27)$$

1.3.3 A set of random variables

The joint pdf of $x_1, \dots, x_n$ satisfies

$$\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} p(x_1, \dots, x_n) dx_1 \dots dx_n = 1. \quad (28)$$

If x_i, x_j are independent, $p(x_i, x_j) = p(x_i) p(x_j)$. Expectation is linear: for $y = \alpha x_i + \beta x_j$,

$$\mathbb{E}(y) = \alpha \mathbb{E}(x_i) + \beta \mathbb{E}(x_j). \quad (29)$$

1.3.4 Covariance

$$\sigma_{ij}^2 = \mathbb{E}((x_i - m_i)(x_j - m_j)) = \mathbb{E}(x_i x_j) - \mathbb{E}(x_i) \mathbb{E}(x_j). \quad (30)$$

When $\sigma_{ij} = 0$ we have $\mathbb{E}(x_i x_j) = \mathbb{E}(x_i) \mathbb{E}(x_j)$ and x_i, x_j are *uncorrelated*.

Discussion questions.

1. If x_i, x_j are independent, then $p(x_i, x_j) = p(x_i)p(x_j) \Rightarrow \mathbb{E}(x_i x_j) = \mathbb{E}(x_i)\mathbb{E}(x_j)$ — so independence implies uncorrelated.
2. If x_i, x_j are uncorrelated, are they necessarily independent?
3. If x_i, x_j are jointly Gaussian, is independence equivalent to being uncorrelated?

1.3.5 Random vector

Let $\mathbf{x} = [x_1, \dots, x_n]^\top$, where all elements are random variables, with mean $\mathbb{E}(\mathbf{x}) \triangleq \mathbf{m}_x$ and covariance

$$\mathbf{P} = \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top) = \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x}) \mathbb{E}(\mathbf{x}^\top), \quad (31)$$

with entries

$$P_{ii} = \mathbb{E}((x_i - m_i)^2) = \int_{-\infty}^{+\infty} (x_i - m_i)^2 p(x_i) dx_i, \quad P_{ij} = \mathbb{E}((x_i - m_i)(x_j - m_j)). \quad (32)$$

Since $P_{ij} = P_{ji}$, the covariance $\mathbf{P}$ is a symmetric matrix; it is also positive (semi-)definite.

1.3.6 Gaussian vector pdf

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} \sqrt{|\mathbf{P}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}^{-1}(\mathbf{x} - \mathbf{m}_x)\right), \quad (33)$$

where $|\mathbf{P}|$ is the determinant of $\mathbf{P}$, $\mathbf{m}_x = \mathbb{E}(\mathbf{x})$, and $\mathbf{P} = \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top)$.

Homework. Let $\mathbf{x} = [x_1, \dots, x_n]^\top$ with $p(\mathbf{x}) = \mathcal{N}(\mathbf{m}_x, \mathbf{P}_{xx})$. Define

$$q = (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}_{xx}^{-1}(\mathbf{x} - \mathbf{m}_x), \quad \mathbf{y} = \mathbf{P}_{xx}^{-1/2}(\mathbf{x} - \mathbf{m}_x), \quad q = \mathbf{y}^\top \mathbf{y} = \sum_i y_i^2.$$

What are the mean and covariance of q ?

1.4 Maximum Likelihood and Maximum a Posteriori Estimation

For a measurement model with additive noise $z_i = h_i(\mathbf{x}) + n_i$, stacked as $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$, two estimators are

- **Maximum likelihood estimation (MLE):** $\max_{\mathbf{x}} p(\mathbf{z} | \mathbf{x})$.
- **Maximum a posteriori (MAP):** $\max_{\mathbf{x}} p(\mathbf{x} | \mathbf{z})$, with $p(\mathbf{x} | \mathbf{z}) = \frac{p(\mathbf{z} | \mathbf{x}) p(\mathbf{x})}{p(\mathbf{z})}$.

For zero-mean Gaussian noise $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{P})$,

$$\max_{\mathbf{x}} p(\mathbf{z} | \mathbf{x}) = \max_{\mathbf{x}} \frac{1}{(2\pi)^{n/2} \sqrt{|\mathbf{P}|}} \exp\left(-(\mathbf{z} - \mathbf{h}(\mathbf{x}))^\top \mathbf{P}^{-1}(\mathbf{z} - \mathbf{h}(\mathbf{x}))\right), \quad (34)$$

which is equivalent to the weighted least-squares problem

$$\min_{\mathbf{x}} C = (\mathbf{z} - \mathbf{h}(\mathbf{x}))^\top \mathbf{P}^{-1}(\mathbf{z} - \mathbf{h}(\mathbf{x})). \quad (35)$$

Linearizing about $\hat{\mathbf{x}}$ with $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n} = \mathbf{h}(\hat{\mathbf{x}}) + \mathbf{H}(\mathbf{x} - \hat{\mathbf{x}}) + \mathbf{n}$, and defining the residual $\tilde{\mathbf{z}} = \mathbf{z} - \mathbf{h}(\hat{\mathbf{x}})$ and error $\tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}$,

$$\min_{\tilde{\mathbf{x}}} (\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}})^\top \mathbf{P}^{-1}(\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}}) = \min_{\tilde{\mathbf{x}}} (\tilde{\mathbf{z}}^\top \tilde{\mathbf{z}} - 2\tilde{\mathbf{z}}^\top \mathbf{P}^{-1}\mathbf{H}\tilde{\mathbf{x}} + \tilde{\mathbf{x}}^\top \mathbf{H}^\top \mathbf{P}^{-1}\mathbf{H}\tilde{\mathbf{x}}). \quad (36)$$

Setting the gradient to zero gives the normal equations $\mathbf{H}^\top \mathbf{P}^{-1}\mathbf{H} \tilde{\mathbf{x}} = \mathbf{H}^\top \mathbf{P}^{-1}\tilde{\mathbf{z}}$, hence

$$\boxed{\mathbf{x}^\oplus = \hat{\mathbf{x}} + \tilde{\mathbf{x}} = \hat{\mathbf{x}} + (\mathbf{H}^\top \mathbf{P}^{-1}\mathbf{H})^{-1} \mathbf{H}^\top \mathbf{P}^{-1}\tilde{\mathbf{z}}.} \quad (37)$$

This is the classical weighted least-squares solution.

Questions.

1. What is the covariance of the new estimate $\mathbf{x}^\oplus$?
2. For a prior $\mathbf{x}_p = \mathbf{x} + \mathbf{n}_p$ together with $z_i = h_i(\mathbf{x}) + n_i$, $i = 1, \dots, n$, what is the MAP solution $\max_{\mathbf{x}} p(\mathbf{x} | \mathbf{z})$?

1.4.1 Optimization

Nonlinear least-squares problems are solved iteratively, e.g. Gauss–Newton (GN) and Levenberg–Marquardt (LM).

1.5 Linear Systems

A linear (continuous-time) state-space system is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}, \quad (38)$$

where $\mathbf{x}$ is the state, $\mathbf{u}$ the input, and $\mathbf{y}$ the output. Its discrete-time counterpart is

$$\mathbf{x}_{k+1} = \mathbf{A}'\mathbf{x}_k + \mathbf{B}'\mathbf{u}_k, \quad \mathbf{y}_{k+1} = \mathbf{C}'\mathbf{x}_{k+1} + \mathbf{D}'\mathbf{u}_{k+1}. \quad (39)$$

If $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are constant, the system is *linear time-invariant* (LTI); if they depend on time, it is *linear time-variant* (LTV).

1.5.1 Homogeneous solution

Consider $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with $\mathbf{x}(t_0) = \mathbf{x}_0$ and $\mathbf{y} = \mathbf{C}\mathbf{x}$. From $\dot{\mathbf{x}} - \mathbf{A}\mathbf{x} = \mathbf{0}$,

$$\frac{d}{dt}(e^{-\mathbf{A}t}\mathbf{x}) = e^{-\mathbf{A}t}\dot{\mathbf{x}} + e^{-\mathbf{A}t}(-\mathbf{A})\mathbf{x} = e^{-\mathbf{A}t}(\dot{\mathbf{x}} - \mathbf{A}\mathbf{x}) = \mathbf{0}. \quad (40)$$

Integrating from 0 to t gives $e^{-\mathbf{A}t}\mathbf{x}(t) - \mathbf{x}(t_0) = \mathbf{0}$, so

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0 = \Phi(t, 0)\mathbf{x}_0, \quad \Phi(t, 0) = e^{\mathbf{A}t}, \quad (41)$$

where $\Phi(t, 0)$ is the *state transition matrix*, and $\mathbf{y} = \mathbf{C}\Phi(t, 0)\mathbf{x}_0$.

1.5.2 Properties of the state transition matrix

$$\Phi(t_2, t_0) = \Phi(t_2, t_1) \Phi(t_1, t_0), \quad \frac{d}{dt} \Phi(t, 0) = \mathbf{A} \Phi(t, 0). \quad (42)$$

The second follows from $\dot{\mathbf{x}} = \dot{\Phi}(t, 0)\mathbf{x}_0 = \mathbf{A}\mathbf{x} = \mathbf{A}\Phi(t, 0)\mathbf{x}_0$.

1.5.3 Output sequence and observability

Sampling the homogeneous response at $t_0, t_1, \dots, t_n$ with $\mathbf{x}_k = \Phi(k, 0)\mathbf{x}_0$ and $\mathbf{y}_k = \mathbf{C}\mathbf{x}_k$ stacks into

$$\begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_n \end{bmatrix} = \begin{bmatrix} \mathbf{C} \Phi(1, 0) \\ \vdots \\ \mathbf{C} \Phi(n, 0) \end{bmatrix} \mathbf{x}_0, \quad (43)$$

which relates the measured outputs to the initial state (an observability-type relation).

1.5.4 Full LTV solution

For the complete system $\dot{\mathbf{x}} = \mathbf{A}(t)\mathbf{x} + \mathbf{B}(t)\mathbf{u}(t)$, write $\dot{\mathbf{x}} - \mathbf{A}(t)\mathbf{x} = \mathbf{B}(t)\mathbf{u}(t)$ and use the integrating factor $e^{-\int_0^t \mathbf{A}(\tau) d\tau}$:

$$\frac{d}{dt} \left(e^{-\int_0^t \mathbf{A}(\tau) d\tau} \mathbf{x} \right) = e^{-\int_0^t \mathbf{A}(\tau) d\tau} (\dot{\mathbf{x}} - \mathbf{A}(t)\mathbf{x}) = e^{-\int_0^t \mathbf{A}(\tau) d\tau} \mathbf{B}(t)\mathbf{u}(t). \quad (44)$$

Integrating,

$$e^{-\int_0^t \mathbf{A}(\tau) d\tau} \mathbf{x}(t) - \mathbf{x}_0 = \int_0^t e^{-\int_0^\tau \mathbf{A}(s) ds} \mathbf{B}(\tau) \mathbf{u}(\tau) d\tau, \quad (45)$$

so that, with $\Phi(t, 0) = e^{\int_0^t \mathbf{A}(\tau) d\tau}$ and $\Phi(t, \tau) = e^{\int_\tau^t \mathbf{A}(s) ds}$,

$$\mathbf{x}(t) = \Phi(t, 0) \mathbf{x}_0 + \int_0^t \Phi(t, \tau) \mathbf{B}(\tau) \mathbf{u}(\tau) d\tau, \quad (46)$$

$$\mathbf{y}(t) = \mathbf{C}(t) \left(\Phi(t, 0) \mathbf{x}_0 + \int_0^t \Phi(t, \tau) \mathbf{B}(\tau) \mathbf{u}(\tau) d\tau \right) + \mathbf{D}(t) \mathbf{u}(t). \quad (47)$$

This is one of the most important properties of linear systems.

1.6 Homework Problems and Solutions

1.6.1 Variance identity

Problem. Show that the variance of a scalar random variable can be written as $\sigma_x^2 = \mathbb{E}(\mathbf{x}^2) - (\mathbb{E}(\mathbf{x}))^2$, and discuss when it vanishes.

Solution. Expanding the definition with $m_x = \mathbb{E}(\mathbf{x})$,

$$\sigma_x^2 = \mathbb{E}((\mathbf{x} - m_x)^2) \quad (48)$$

$$= \mathbb{E}(\mathbf{x}^2 - 2m_x \mathbf{x} + m_x^2) \quad (49)$$

$$= \mathbb{E}(\mathbf{x}^2) - 2m_x \mathbb{E}(\mathbf{x}) + m_x^2 \quad (50)$$

$$= \mathbb{E}(\mathbf{x}^2) - m_x^2. \quad (51)$$

Since $\sigma_x^2 \geq 0$, we have $\mathbb{E}(\mathbf{x}^2) \geq (\mathbb{E}(\mathbf{x}))^2$, with equality if and only if $\sigma_x^2 = 0$, i.e. $\mathbf{x} = m_x$ almost surely (a deterministic variable) — a degenerate special case.

1.6.2 Covariance matrix identity

Problem. For a random vector $\mathbf{x}$ with mean $\mathbf{m}_x$, show that $\mathbf{P} = \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top) = \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top)$.

Solution. Expanding the outer product and using linearity of expectation, together with $\mathbb{E}(\mathbf{x}) = \mathbf{m}_x$,

$$\mathbf{P} = \mathbb{E}(\mathbf{x}\mathbf{x}^\top - \mathbf{x}\mathbf{m}_x^\top - \mathbf{m}_x\mathbf{x}^\top + \mathbf{m}_x\mathbf{m}_x^\top) \quad (52)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbf{m}_x^\top - \mathbf{m}_x\mathbb{E}(\mathbf{x}^\top) + \mathbf{m}_x\mathbf{m}_x^\top \quad (53)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbf{m}_x\mathbf{m}_x^\top - \mathbf{m}_x\mathbf{m}_x^\top + \mathbf{m}_x\mathbf{m}_x^\top \quad (54)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbf{m}_x\mathbf{m}_x^\top \quad (55)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top). \quad (56)$$

1.6.3 Scalar covariance identity

Problem. For two random variables x_i, x_j , show that $\sigma_{ij} = \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j)$.

Solution. With $m_i = \mathbb{E}(x_i)$, $m_j = \mathbb{E}(x_j)$, expand

$$\sigma_{ij} = \mathbb{E}((x_i - m_i)(x_j - m_j)) \quad (57)$$

$$= \iint (x_i - m_i)(x_j - m_j) p(x_i, x_j) dx_i dx_j \quad (58)$$

$$\begin{aligned} &= \iint x_i x_j p(x_i, x_j) dx_i dx_j - m_j \iint x_i p(x_i, x_j) dx_i dx_j \\ &\quad - m_i \iint x_j p(x_i, x_j) dx_i dx_j + m_i m_j \iint p(x_i, x_j) dx_i dx_j. \end{aligned} \quad (59)$$

Using the marginalization $\int p(x_i, x_j) dx_i = p(x_j)$ (and symmetrically), together with $\iint p(x_i, x_j) dx_i dx_j = 1$,

$$m_i \iint x_j p(x_i, x_j) dx_i dx_j = m_i \int x_j p(x_j) dx_j = m_i m_j, \quad (60)$$

and likewise for the other term, so

$$\sigma_{ij} = \mathbb{E}(x_i x_j) - m_i m_j - m_i m_j + m_i m_j \quad (61)$$

$$= \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j). \quad (62)$$

1.6.4 Independence versus uncorrelatedness

Problem. Resolve the three discussion questions: (1) does independence imply uncorrelatedness? (2) does uncorrelatedness imply independence? (3) for jointly Gaussian variables, are the two equivalent?

Solution. (1) *Independent $\Rightarrow$ uncorrelated.* If $p(x_i, x_j) = p(x_i)p(x_j)$, then

$$\mathbb{E}(x_i x_j) = \iint x_i x_j p(x_i, x_j) dx_i dx_j \quad (63)$$

$$= \int x_i p(x_i) dx_i \cdot \int x_j p(x_j) dx_j \quad (64)$$

$$= \mathbb{E}(x_i)\mathbb{E}(x_j), \quad (65)$$

so $\sigma_{ij} = 0$.

(2) *Uncorrelated $\not\Rightarrow$ independent.* Uncorrelatedness gives only

$$\mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j) = \iint x_i x_j (p(x_i, x_j) - p(x_i)p(x_j)) dx_i dx_j = 0, \quad (66)$$

which does *not* force $p(x_i, x_j) = p(x_i)p(x_j)$ in general. Hence uncorrelated does not imply independent.

(3) *Jointly Gaussian.* For $x_i \sim \mathcal{N}(m_i, \sigma_i^2)$ and $x_j \sim \mathcal{N}(m_j, \sigma_j^2)$, the product of marginals is

$$p(x_i)p(x_j) = \frac{1}{2\pi\sigma_i\sigma_j} \exp\left\{-\frac{1}{2} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^\top \begin{bmatrix} \sigma_i^2 & 0 \\ 0 & \sigma_j^2 \end{bmatrix}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}\right\}. \quad (67)$$

The joint Gaussian $[x_i, x_j]^\top \sim \mathcal{N}([m_i, m_j]^\top, \mathbf{P}_{xx})$ with $\mathbf{P}_{xx} = \begin{bmatrix} \sigma_i^2 & \sigma_{ij} \\ \sigma_{ji} & \sigma_j^2 \end{bmatrix}$ is

$$p(x_i, x_j) = \frac{1}{2\pi\sqrt{|\mathbf{P}_{xx}|}} \exp\left\{-\frac{1}{2} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^\top \mathbf{P}_{xx}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}\right\}. \quad (68)$$

If $\sigma_{ij} = \sigma_{ji} = 0$, then $\mathbf{P}_{xx}$ is diagonal and $p(x_i, x_j) = p(x_i)p(x_j)$. Hence, for (jointly) Gaussian variables, *independent $\Leftrightarrow$ uncorrelated*.

1.6.5 The χ^2 distribution

Problem. Let $\mathbf{x} = [x_1, \dots, x_n]^\top$ with $p(\mathbf{x}) = \mathcal{N}(\mathbf{m}_x, \mathbf{P}_{xx})$ and define $q = (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}_{xx}^{-1} (\mathbf{x} - \mathbf{m}_x)$. Find the mean and variance of q .

Solution. Factor $\mathbf{P}_{xx} = \mathbf{L}\mathbf{L}^\top$ (Cholesky), so $\mathbf{P}_{xx}^{-1} = \mathbf{L}^{-\top}\mathbf{L}^{-1}$, and define the whitened variable $\mathbf{y} = \mathbf{L}^{-1}(\mathbf{x} - \mathbf{m}_x)$. Then

$$q = (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{L}^{-\top} \mathbf{L}^{-1} (\mathbf{x} - \mathbf{m}_x) \quad (69)$$

$$= \mathbf{y}^\top \mathbf{y} = \sum_{i=1}^n y_i^2. \quad (70)$$

The whitened variable has zero mean and identity covariance,

$$\mathbf{m}_y = \mathbb{E}(\mathbf{y}) = \mathbf{L}^{-1}(\mathbb{E}(\mathbf{x}) - \mathbf{m}_x) = \mathbf{0}, \quad (71)$$

$$\mathbf{P}_{yy} = \mathbb{E}(\mathbf{y}\mathbf{y}^\top) \quad (72)$$

$$= \mathbf{L}^{-1} \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top) \mathbf{L}^{-\top} \quad (73)$$

$$= \mathbf{L}^{-1} \mathbf{P}_{xx} \mathbf{L}^{-\top} \quad (74)$$

$$= \mathbf{L}^{-1} \mathbf{L} \mathbf{L}^\top \mathbf{L}^{-\top} = \mathbf{I}_n, \quad (75)$$

so $y_i \sim \mathcal{N}(0, 1)$ are independent. Therefore the mean of q is

$$\mathbb{E}(q) = \mathbb{E}(\mathbf{y}^\top \mathbf{y}) = \sum_{i=1}^n \mathbb{E}(y_i^2) = n. \quad (76)$$

For the variance, with $\mathbb{E}(y_i^2) = 1$ the cross terms vanish by independence, giving

$$\sigma_{qq}^2 = \mathbb{E}((q - n)^2) = \mathbb{E}\left(\left(\sum_i (y_i^2 - 1)\right)^2\right) \quad (77)$$

$$= \sum_i \mathbb{E}((y_i^2 - 1)^2) \quad (78)$$

$$= \sum_i (\mathbb{E}(y_i^4) - 2\mathbb{E}(y_i^2) + 1). \quad (79)$$

The fourth moment of a standard normal follows from integration by parts (with $u = y_i^3$, $dv = y_i p(y_i) dy_i$, $v = -p(y_i)$):

$$\mathbb{E}(y_i^4) = \int_{-\infty}^{+\infty} y_i^4 p(y_i) dy_i \quad (80)$$

$$= \underbrace{[-y_i^3 p(y_i)]_{-\infty}^{+\infty}}_{=0} + \int_{-\infty}^{+\infty} 3y_i^2 p(y_i) dy_i \quad (81)$$

$$= 3\mathbb{E}(y_i^2) = 3. \quad (82)$$

Hence each term equals $3 - 2 + 1 = 2$, and

$$\sigma_{qq}^2 = \sum_{i=1}^n 2 = 2n. \quad (83)$$

So $q \sim \chi^2(n)$ has mean n and variance $2n$.

2 Lecture 2: Rotation, Lie Groups, SE(3), SE₂(3), and Quaternions

Further reading. This lecture draws on several standard references: matrix Lie groups in Barfoot [1, Ch. 7]; the Lie-group treatment of $SO(3)/SE(3)$ in the GTSAM notes [2] and Atanasov's ECE276A slides [3]; and quaternion algebra/kinematics in Trawny & Roumeliotis [4, Sec. 1] and Solà [5, Sec. 1–3].

2.1 Preliminaries: useful series

The matrix exponential and the sine/cosine series underpin everything that follows:

$$e^x = \exp(x) = \sum_{k=0}^{\infty} \frac{1}{k!} x^k = I + x + \frac{1}{2!} x^2 + \frac{1}{3!} x^3 + \frac{1}{4!} x^4 + \dots, \quad (84)$$

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} = x - \frac{1}{3!} x^3 + \frac{1}{5!} x^5 - \dots, \quad (85)$$

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} = 1 - \frac{1}{2!} x^2 + \frac{1}{4!} x^4 - \dots. \quad (86)$$

For small x (so that $x^2, x^3 \rightarrow 0$), the important first-order approximations are $e^x \approx I + x$, $\sin(x) \approx x$, and $\cos(x) \approx 1$.

2.2 Frames, notation, and transformations

We work with coordinate frames $\{A\}$ and $\{B\}$. A point/vector $\mathbf{L}$ expressed in $\{A\}$ is ${}^A\mathbf{L}$, and in $\{B\}$ is ${}^B\mathbf{L}$. The angle $\theta_{A \rightarrow B}$ is the rotation about the z -axis taking $\{A\}$ to $\{B\}$ (Fig. 1).

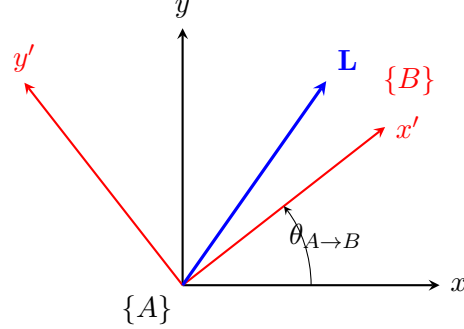


Figure 1: Frame $\{B\}$ is frame $\{A\}$ rotated by $\theta_{A \rightarrow B}$ about the z -axis; the vector $\mathbf{L}$ can be expressed in either frame.

The rotation matrix ${}^A_B\mathbf{R}$ maps a vector from $\{B\}$ to $\{A\}$:

$${}^A\mathbf{L} = {}^A_B\mathbf{R} {}^B\mathbf{L} = {}^A_B\mathbf{R}(\theta_{A \rightarrow B}) {}^B\mathbf{L}. \quad (87)$$

Adding the translation ${}^A\mathbf{p}_B$ (the origin of $\{B\}$ expressed in $\{A\}$, Fig. 2) gives the full rigid-body transformation

$$\{A\}, \{B\} \Rightarrow \{{}^A_B\mathbf{R}, {}^A\mathbf{p}_B\} \Rightarrow {}^A_B\mathbf{T} = \begin{bmatrix} {}^A_B\mathbf{R} & {}^A\mathbf{p}_B \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}. \quad (88)$$

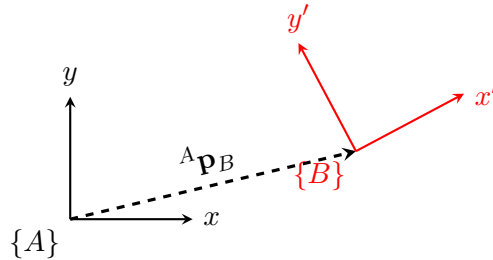


Figure 2: A rigid-body transformation combines the rotation ${}^A_B\mathbf{R}$ with the translation ${}^A\mathbf{p}_B$.

2.3 Defining rotation

2.3.1 Euler angles

With roll θ (about x), pitch r (about y), yaw ψ (about z),

$$\mathbf{R}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}, \quad \mathbf{R}(r) = \begin{bmatrix} \cos r & 0 & \sin r \\ 0 & 1 & 0 \\ -\sin r & 0 & \cos r \end{bmatrix}, \quad \mathbf{R}(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (89)$$

and the yaw–pitch–roll composition is

$$\mathbf{R} = \mathbf{R}(\psi) \mathbf{R}(r) \mathbf{R}(\theta) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos r & 0 & \sin r \\ 0 & 1 & 0 \\ -\sin r & 0 & \cos r \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}. \quad (90)$$

2.3.2 Infinitesimal rotation and the skew-symmetric matrix

For small angles $\delta\theta, \delta r, \delta\psi \rightarrow 0$ each elementary rotation linearizes ($\cos \rightarrow 1$, $\sin \rightarrow \text{angle}$):

$$\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\theta \\ 0 & \delta\theta & 1 \end{bmatrix}, \quad \mathbf{R}(\delta r) = \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & 0 \\ -\delta r & 0 & 1 \end{bmatrix}, \quad \mathbf{R}(\delta\psi) = \begin{bmatrix} 1 & -\delta\psi & 0 \\ \delta\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (91)$$

Multiplying the last two and dropping all second-order products ($\delta r \delta\theta \approx 0$ etc.),

$$\mathbf{R}(\delta r)\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & 0 \\ -\delta r & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\theta \\ 0 & \delta\theta & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix}, \quad (92)$$

and then

$$\begin{aligned} \mathbf{R} &= \mathbf{R}(\delta\psi)\mathbf{R}(\delta r)\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & -\delta\psi & 0 \\ \delta\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -\delta\psi & \delta r \\ \delta\psi & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix} = \mathbf{I} + \begin{bmatrix} 0 & -\delta\psi & \delta r \\ \delta\psi & 0 & -\delta\theta \\ -\delta r & \delta\theta & 0 \end{bmatrix} = \mathbf{I} + [\boldsymbol{\theta} \times], \end{aligned} \quad (93)$$

where the small-angle vector is $\boldsymbol{\theta} = [\delta\theta, \delta r, \delta\psi]^\top$. The matrix $[\boldsymbol{\theta} \times]$ is skew-symmetric, $[\boldsymbol{\theta} \times] = -[\boldsymbol{\theta} \times]^\top$. This motivates writing a finite rotation as $\mathbf{R}(\boldsymbol{\theta}) \approx \mathbf{I} + [\boldsymbol{\theta} \times] \approx \exp([\boldsymbol{\theta} \times])$, and defining $\text{Exp}(\boldsymbol{\theta}) \triangleq \exp([\boldsymbol{\theta} \times])$. The question is then whether $\mathbf{R}(\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta})$.

2.4 Axis-angle and Rodrigues' formula

Let $\mathbf{k}$ be the unit rotation axis and θ the rotation angle, with $\boldsymbol{\theta} = \theta\mathbf{k} = [\theta_1, \theta_2, \theta_3]^\top$, so

$$\mathbf{k} = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix}, \quad [\mathbf{k} \times] = \begin{bmatrix} 0 & -k_3 & k_2 \\ k_3 & 0 & -k_1 \\ -k_2 & k_1 & 0 \end{bmatrix}, \quad [\boldsymbol{\theta} \times] = \begin{bmatrix} 0 & -\theta_3 & \theta_2 \\ \theta_3 & 0 & -\theta_1 \\ -\theta_2 & \theta_1 & 0 \end{bmatrix}. \quad (94)$$

The skew operator on a unit vector satisfies the cyclic power identities

$$[\mathbf{k} \times]^2 = \mathbf{k}\mathbf{k}^\top - \mathbf{I}_3, \quad [\mathbf{k} \times]^3 = -[\mathbf{k} \times], \quad [\mathbf{k} \times]^4 = -[\mathbf{k} \times]^2, \quad [\mathbf{k} \times]^5 = [\mathbf{k} \times], \quad [\mathbf{k} \times]^6 = [\mathbf{k} \times]^2, \quad [\mathbf{k} \times]^7 = -[\mathbf{k} \times] \quad (95)$$

Substituting $[\boldsymbol{\theta} \times] = \theta [\mathbf{k} \times]$ into the exponential series term by term,

$$\begin{aligned} \text{Exp}(\boldsymbol{\theta}) &= \exp(\theta [\mathbf{k} \times]) = \mathbf{I} + \theta [\mathbf{k} \times] + \frac{\theta^2}{2!} [\mathbf{k} \times]^2 + \frac{\theta^3}{3!} [\mathbf{k} \times]^3 + \frac{\theta^4}{4!} [\mathbf{k} \times]^4 + \frac{\theta^5}{5!} [\mathbf{k} \times]^5 + \frac{\theta^6}{6!} [\mathbf{k} \times]^6 + \frac{\theta^7}{7!} [\mathbf{k} \times]^7 + \dots \\ &= \mathbf{I} + \theta [\mathbf{k} \times] + \frac{\theta^2}{2!} [\mathbf{k} \times]^2 - \frac{\theta^3}{3!} [\mathbf{k} \times] - \frac{\theta^4}{4!} [\mathbf{k} \times]^2 + \frac{\theta^5}{5!} [\mathbf{k} \times] + \frac{\theta^6}{6!} [\mathbf{k} \times]^2 - \frac{\theta^7}{7!} [\mathbf{k} \times] + \dots \\ &= \mathbf{I} + \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots\right)}_{\sin \theta} [\mathbf{k} \times] + \underbrace{\left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \dots\right)}_{1 - \cos \theta} [\mathbf{k} \times]^2, \end{aligned} \quad (96)$$

which gives **Rodrigues' rotation formula**

$$\boxed{\mathbf{R}(\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) = \exp(\theta [\mathbf{k} \times]) = \mathbf{I} + \sin \theta [\mathbf{k} \times] + (1 - \cos \theta) [\mathbf{k} \times]^2.} \quad (97)$$

For a small angle $\delta\theta \rightarrow 0$, $\mathbf{R}(\delta\boldsymbol{\theta}) \approx \mathbf{I} + [\delta\boldsymbol{\theta} \times]$ (small-angle approximation).

2.4.1 Logarithmic map

The log map is the inverse of the exponential map:

$$\log(\mathbf{R}) = \lfloor \boldsymbol{\theta} \times \rfloor, \quad \text{Log}(\mathbf{R}) = \boldsymbol{\theta}_{3 \times 1}, \quad \theta = \arccos\left(\frac{\text{tr} \mathbf{R} - 1}{2}\right), \quad \lfloor \mathbf{k} \times \rfloor = \frac{\mathbf{R} - \mathbf{R}^\top}{2 \sin \theta}. \quad (98)$$

2.5 SO(3) kinematics

The two directions of the maps are: from the vector space $\mathbb{R}^3$ via the Lie algebra to the Lie group, $\boldsymbol{\theta} \rightarrow \lfloor \boldsymbol{\theta} \times \rfloor \rightarrow \mathbf{R} = \text{Exp}(\boldsymbol{\theta})$; and back, $\mathbf{R} \rightarrow \lfloor \boldsymbol{\theta} \times \rfloor \rightarrow \boldsymbol{\theta} = \text{Log}(\mathbf{R})$.

(1) Angular-velocity kinematics. Differentiating the orthogonality constraint $\mathbf{R}\mathbf{R}^\top = \mathbf{I}_3$,

$$\frac{d}{dt}(\mathbf{R}\mathbf{R}^\top) = \mathbf{0} \Rightarrow \dot{\mathbf{R}}\mathbf{R}^\top + \mathbf{R}\dot{\mathbf{R}}^\top = \mathbf{0} \Rightarrow \dot{\mathbf{R}}\mathbf{R}^\top = -\mathbf{R}\dot{\mathbf{R}}^\top = -(\dot{\mathbf{R}}\mathbf{R}^\top)^\top, \quad (99)$$

so $\dot{\mathbf{R}}\mathbf{R}^\top$ is skew-symmetric. Writing it as $\lfloor \boldsymbol{\omega} \times \rfloor$ with $\lfloor \boldsymbol{\omega} \times \rfloor = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$,

$$\dot{\mathbf{R}}\mathbf{R}^\top = \lfloor \boldsymbol{\omega} \times \rfloor \Rightarrow \dot{\mathbf{R}} = \lfloor \boldsymbol{\omega} \times \rfloor \mathbf{R}. \quad (100)$$

(2) Equivalent limit form. With $\mathbf{R}(t + \Delta t) = \mathbf{R}(t) \exp(\lfloor \boldsymbol{\omega} \times \rfloor \Delta t)$,

$$\dot{\mathbf{R}}(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{R}(t + \Delta t) - \mathbf{R}(t)}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{R}(t) \exp(\lfloor \boldsymbol{\omega} \times \rfloor \Delta t) - \mathbf{R}(t)}{\Delta t} = \mathbf{R}(t) \lim_{\Delta t \rightarrow 0} \frac{\exp(\lfloor \boldsymbol{\omega} \times \rfloor \Delta t) - \mathbf{I}}{\Delta t} = \mathbf{R}(t) \lfloor \boldsymbol{\omega} \times \rfloor. \quad (101)$$

(3) Integration. With $\mathbf{R}_i = \exp(\lfloor \boldsymbol{\omega}_i \times \rfloor \Delta t_i)$, composing increments gives

$$\mathbf{R}_{1 \rightarrow k} = \mathbf{R}_k \mathbf{R}_{k-1} \cdots \mathbf{R}_2 \mathbf{R}_1 = \exp(\lfloor \boldsymbol{\omega}_k \times \rfloor \Delta t_k) \cdots \exp(\lfloor \boldsymbol{\omega}_1 \times \rfloor \Delta t_1). \quad (102)$$

(4) Interpolation. For ${}^G_{t_1}\mathbf{R}$, ${}^G_{t_2}\mathbf{R}$ and $t \in [t_1, t_2]$, define the relative rotation and its angle, and the interpolation ratio:

$${}^{t_1}_{t_2}\mathbf{R} = {}^G_{t_1}\mathbf{R}^\top {}^G_{t_2}\mathbf{R}, \quad {}^{t_1}_{t_2}\boldsymbol{\theta} = \text{Log}({}^{t_1}_{t_2}\mathbf{R}) = \text{Log}({}^G_{t_1}\mathbf{R}^\top {}^G_{t_2}\mathbf{R}), \quad \lambda = \frac{t - t_1}{t_2 - t_1}. \quad (103)$$

Then

$${}^G_t\mathbf{R} = {}^G_{t_1}\mathbf{R} {}^{t_1}_t\mathbf{R} = {}^G_{t_1}\mathbf{R} \text{Exp}(\lambda {}^{t_1}_{t_2}\boldsymbol{\theta}). \quad (104)$$

(5) Right Jacobian. For a rotation $\mathbf{R}(\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta})$, perturb the angle by $\delta\boldsymbol{\theta} \in \mathbb{R}^3$ and let $\delta\boldsymbol{\varphi}$ be the corresponding small rotation in $\mathfrak{so}(3)$:

$$\mathbf{R}(\boldsymbol{\theta}) \text{Exp}(\delta\boldsymbol{\varphi}) = \mathbf{R}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \Rightarrow \text{Exp}(\boldsymbol{\theta}) \text{Exp}(\delta\boldsymbol{\varphi}) = \text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}). \quad (105)$$

Solving for $\delta\boldsymbol{\varphi}$,

$$\text{Exp}(\delta\boldsymbol{\varphi}) = \text{Exp}(\boldsymbol{\theta})^{-1} \text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \Rightarrow \delta\boldsymbol{\varphi} = \text{Log}(\text{Exp}(\boldsymbol{\theta})^{-1} \text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta})), \quad (106)$$

and the *right Jacobian* is the derivative of $\delta\varphi$ with respect to $\delta\theta$:

$$\mathbf{J}_r(\theta) \triangleq \frac{\partial \delta\varphi}{\partial \delta\theta} = \lim_{\delta\theta \rightarrow 0} \frac{\text{Log}(\text{Exp}(\theta)^{-1} \text{Exp}(\theta + \delta\theta))}{\delta\theta}. \quad (107)$$

Equivalently, $\text{Exp}(\theta + \delta\theta) = \text{Exp}(\theta) \text{Exp}(\mathbf{J}_r(\theta) \delta\theta)$, and inverting,

$$\text{Exp}(\theta) \text{Exp}(\delta\varphi) = \text{Exp}(\theta + \mathbf{J}_r^{-1}(\theta) \delta\varphi) \Rightarrow \text{Log}(\text{Exp}(\theta) \text{Exp}(\delta\varphi)) = \theta + \mathbf{J}_r^{-1}(\theta) \delta\varphi, \quad (108)$$

with the closed form

$$\mathbf{J}_r(\theta) = \mathbf{I} - \frac{1 - \cos\|\theta\|}{\|\theta\|^2} [\theta \times] + \frac{\|\theta\| - \sin\|\theta\|}{\|\theta\|^3} [\theta \times]^2. \quad (109)$$

(6) Left Jacobian. Applying the perturbation on the left instead defines the left Jacobian:

$$\text{Exp}(\theta + \delta\theta) = \text{Exp}(\theta) \text{Exp}(\mathbf{J}_r(\theta) \delta\theta) = \text{Exp}(\mathbf{J}_l(\theta) \delta\theta) \text{Exp}(\theta). \quad (110)$$

Using the adjoint identity $\text{Exp}(\theta) \text{Exp}(\mathbf{x}) \text{Exp}(\theta)^{-1} = \text{Exp}(\mathbf{R}(\theta)\mathbf{x})$ on the middle expression,

$$\text{Exp}(\theta) \text{Exp}(\mathbf{J}_r \delta\theta) = \text{Exp}(\mathbf{R}(\theta) \mathbf{J}_r \delta\theta) \text{Exp}(\theta) \Rightarrow \mathbf{J}_l(\theta) = \mathbf{R}(\theta) \mathbf{J}_r(\theta). \quad (111)$$

(7) Sign symmetry. From $\text{Exp}(\theta + \delta\theta) = \text{Exp}(\theta) \text{Exp}(\mathbf{J}_r(\theta) \delta\theta)$, replacing $\theta \rightarrow -\theta$, $\delta\theta \rightarrow -\delta\theta$,

$$\text{Exp}(-\theta - \delta\theta) = \text{Exp}(-\mathbf{J}_l(-\theta) \delta\theta) \text{Exp}(-\theta) = \text{Exp}(-\mathbf{J}_r(\theta) \delta\theta) \text{Exp}(-\theta) \Rightarrow \mathbf{J}_l(-\theta) = \mathbf{J}_r(\theta). \quad (112)$$

(8) Summary. Collecting (5)–(7),

$$\text{Exp}(\theta + \delta\theta) = \text{Exp}(\theta) \text{Exp}(\mathbf{J}_r(\theta) \delta\theta), \quad \mathbf{J}_l(\theta) = \mathbf{R}(\theta) \mathbf{J}_r(\theta), \quad \mathbf{J}_l(-\theta) = \mathbf{J}_r(\theta). \quad (113)$$

(9) Identity $[\mathbf{R}\omega \times] = \mathbf{R} [\omega \times] \mathbf{R}^\top$. For any vector $\mathbf{v}$, using $\mathbf{R}(\mathbf{u} \times \mathbf{v}) = (\mathbf{R}\mathbf{u}) \times (\mathbf{R}\mathbf{v})$ and $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$,

$$[\mathbf{R}\mathbf{u} \times] (\mathbf{R}\mathbf{v}) = \mathbf{R} ([\mathbf{u} \times] \mathbf{v}) \quad (114)$$

$$= \mathbf{R} ([\mathbf{u} \times] \mathbf{R}^\top \mathbf{R}\mathbf{v}) \quad (115)$$

$$= \mathbf{R} [\mathbf{u} \times] \mathbf{R}^\top (\mathbf{R}\mathbf{v}). \quad (116)$$

Since this holds for all $\mathbf{R}\mathbf{v}$, we conclude $[\mathbf{R}\mathbf{u} \times] = \mathbf{R} [\mathbf{u} \times] \mathbf{R}^\top$.

Homework. Prove $[\mathbf{R}\omega \times] = \mathbf{R} [\omega \times] \mathbf{R}^\top$.

(10) Identity $\text{Exp}(\mathbf{R}\varphi) = \mathbf{R} \text{Exp}(\varphi) \mathbf{R}^\top$. Using $[\mathbf{R}\varphi \times]^k = \mathbf{R} [\varphi \times]^k \mathbf{R}^\top$ (which follows from (9), since $[\mathbf{R}\varphi \times]^2 = \mathbf{R} [\varphi \times] \mathbf{R}^\top \mathbf{R} [\varphi \times] \mathbf{R}^\top = \mathbf{R} [\varphi \times]^2 \mathbf{R}^\top$, etc.),

$$\begin{aligned} \text{Exp}(\mathbf{R}\varphi) &= \mathbf{I} + [\mathbf{R}\varphi \times] + \frac{1}{2!} [\mathbf{R}\varphi \times]^2 + \frac{1}{3!} [\mathbf{R}\varphi \times]^3 + \cdots \\ &= \mathbf{R}\mathbf{R}^\top + \mathbf{R} [\varphi \times] \mathbf{R}^\top + \frac{1}{2!} \mathbf{R} [\varphi \times]^2 \mathbf{R}^\top + \cdots \\ &= \mathbf{R} (\mathbf{I} + [\varphi \times] + \frac{1}{2!} [\varphi \times]^2 + \frac{1}{3!} [\varphi \times]^3 + \cdots) \mathbf{R}^\top = \mathbf{R} \text{Exp}(\varphi) \mathbf{R}^\top. \end{aligned} \quad (117)$$

Homework. Prove $\text{Exp}(\mathbf{R}\varphi) = \mathbf{R} \text{Exp}(\varphi) \mathbf{R}^\top$.

2.5.1 Worked applications

(11) Recovering ${}^C_I\mathbf{R}$ from vector pairs. Given camera frame $\{C\}$ and IMU frame $\{I\}$ with ${}^C\mathbf{v}_1 = {}^C_I\mathbf{R} {}^I\mathbf{v}_1$ and ${}^C\mathbf{v}_2 = {}^C_I\mathbf{R} {}^I\mathbf{v}_2$, the trick is to stack three independent directions (the third from the cross product) into an invertible matrix:

$$\begin{bmatrix} {}^C\mathbf{v}_1 & {}^C\mathbf{v}_2 & {}^C\mathbf{v}_1 \times {}^C\mathbf{v}_2 \end{bmatrix} = {}^C_I\mathbf{R} \begin{bmatrix} {}^I\mathbf{v}_1 & {}^I\mathbf{v}_2 & {}^I\mathbf{v}_1 \times {}^I\mathbf{v}_2 \end{bmatrix}. \quad (118)$$

(12) Given instead relative rotations ${}^{C_1}_{C_2}\mathbf{R} = {}^C_I\mathbf{R} {}^{I_1}_{I_2}\mathbf{R} {}^C_I\mathbf{R}^\top$, recover ${}^C_I\mathbf{R}$ from two such pairs.

Discussion. For (11)/(12): What is the minimum number of data pairs needed? What requirements must the pairs satisfy? What if more than the minimum is available?

(14) Averaging noisy rotations. Given n measurements $\mathbf{R}_{m_i} = \mathbf{R} \exp([\mathbf{n}_i \times])$ with $\mathbf{n}_i \sim \mathcal{N}(\mathbf{0}_{3 \times 1}, \mathbf{P}_i)$, estimate $\mathbf{R}$ and its covariance.

2.6 Groups: $SO(3)$ and $SE(3)$

$SO(3)$ — special orthogonal group.

$$SO(3) = \{\mathbf{R} \in \mathbb{R}^{3 \times 3} \mid \mathbf{R}^\top \mathbf{R} = \mathbf{I}_3, \det(\mathbf{R}) = 1\}. \quad (119)$$

It satisfies the four group axioms: *closure* ($\mathbf{R}_1\mathbf{R}_2 \in SO(3)$), *identity* ($\mathbf{I}_3 \in SO(3)$), *inverse* ($\mathbf{R}^{-1} = \mathbf{R}^\top \in SO(3)$), and *associativity* ($(\mathbf{R}_1\mathbf{R}_2)\mathbf{R}_3 = \mathbf{R}_1(\mathbf{R}_2\mathbf{R}_3)$).

$SE(3)$ — special Euclidean group.

$$SE(3) = \left\{ \mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \mid \mathbf{R} \in SO(3), \mathbf{p} \in \mathbb{R}^3 \right\}. \quad (120)$$

The four axioms: *closure*

$$\mathbf{T}_1\mathbf{T}_2 = \begin{bmatrix} \mathbf{R}_1 & \mathbf{p}_1 \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_2 & \mathbf{p}_2 \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_1\mathbf{R}_2 & \mathbf{R}_1\mathbf{p}_2 + \mathbf{p}_1 \\ \mathbf{0} & 1 \end{bmatrix} \in SE(3), \quad (121)$$

identity $\begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}$, *inverse*

$$\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{R}^\top & -\mathbf{R}^\top \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \in SE(3), \quad (122)$$

and *associativity* $(\mathbf{T}_1\mathbf{T}_2)\mathbf{T}_3 = \mathbf{T}_1(\mathbf{T}_2\mathbf{T}_3)$.

2.6.1 Matrix Lie groups

A *group* is a set with an operation satisfying closure, associativity, identity, and invertibility. A *Lie group* is a group that is also a smooth (differentiable) manifold whose group operation is smooth. A *matrix Lie group* has matrices as its elements. The maps between the vector space, the Lie algebra, and the Lie group are

$$\mathbb{R}^m \xrightarrow{\wedge \text{ (wedge)}} \mathfrak{so}(n) \xrightarrow{\exp} SO(n), \quad SO(n) \xrightarrow{\log} \mathfrak{so}(n) \xrightarrow{\vee \text{ (vee)}} \mathbb{R}^m, \quad (123)$$

and for short we write $\mathbf{X} = \text{Exp}(\mathbf{x}) = \exp(\mathbf{x}^\wedge)$ and $\mathbf{x} = \text{Log}(\mathbf{X}) = \log(\mathbf{X})^\vee$. For $SO(3)$ specifically, $\boldsymbol{\theta} \in \mathbb{R}^3 \rightarrow [\boldsymbol{\theta} \times] = \boldsymbol{\theta}^\wedge \rightarrow \mathbf{R} = \exp(\boldsymbol{\theta}^\wedge)$, and $\mathbf{R} \in SO(3) \rightarrow [\boldsymbol{\theta} \times] = \log(\mathbf{R}) \rightarrow \boldsymbol{\theta} = \log(\mathbf{R})^\vee$.

2.6.2 Why this (manifold) machinery?

It *simplifies* estimation: the core idea is to use a smooth manifold to model parameters that are awkward in plain vector space, which makes optimization, interpolation, integration, and differentiation easier or even possible (Fig. 3). For example, a raw angle $\theta \in [-\pi, \pi]$ has a wrap-around cut; the circle/sphere is smooth, and the tangent line at a point models the small change / error state used in optimization.

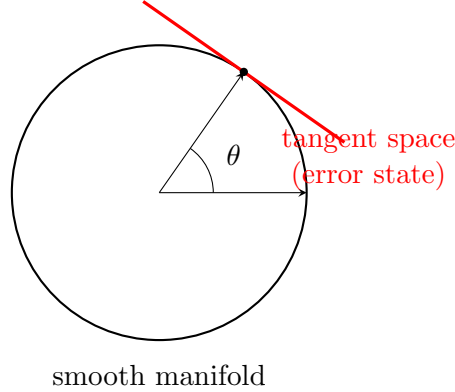


Figure 3: A smooth manifold (here a circle/sphere) avoids the wrap-around of a raw angle; the red tangent line is the local vector space used to model derivatives and error states for optimization.

For $SO(3)$ the left Jacobian also appears directly in the exponential expansion:

$$\mathbf{R}(\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) = \mathbf{I}_3 + [\boldsymbol{\theta} \times] \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{I}_3 + \boldsymbol{\theta}^\wedge \mathbf{J}_l(\boldsymbol{\theta}), \quad \mathbf{J}_l(\boldsymbol{\theta}) = \sum_{n=0}^{\infty} \frac{1}{(n+1)!} (\boldsymbol{\theta}^\wedge)^n. \quad (124)$$

2.7 SE(3)

Exp / Log. The $SE(3)$ tangent vector uses $\boldsymbol{\rho}$ for its translation component, $\boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \end{bmatrix} \in \mathbb{R}^6$, which is *distinct* from the group translation $\mathbf{p}$ in $\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}$; the two are related by $\mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta}) \boldsymbol{\rho}$. With the 4×4 algebra element $\boldsymbol{\xi}^\wedge = \begin{bmatrix} [\boldsymbol{\theta} \times] & \boldsymbol{\rho} \\ \mathbf{0}_{1 \times 3} & 0 \end{bmatrix}$,

$$\mathbf{T} = \text{Exp}(\boldsymbol{\xi}) = \exp \begin{bmatrix} [\boldsymbol{\theta} \times] & \boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} = \begin{bmatrix} \exp([\boldsymbol{\theta} \times]) & \mathbf{J}_l(\boldsymbol{\theta}) \boldsymbol{\rho} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}, \quad \mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta}) \boldsymbol{\rho}, \quad (125)$$

and inverting,

$$\boldsymbol{\xi} = \text{Log}(\mathbf{T})^\vee, \quad \log(\mathbf{T}) = \log \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} \log(\mathbf{R}) & \mathbf{J}_l^{-1}(\boldsymbol{\theta}) \mathbf{p} \\ \mathbf{0} & 0 \end{bmatrix} \Rightarrow \boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\theta} \\ \mathbf{J}_l^{-1}(\boldsymbol{\theta}) \mathbf{p} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \end{bmatrix}. \quad (126)$$

Discussion. Prove $\exp \left(\begin{bmatrix} [\boldsymbol{\theta} \times] & \boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \right) = \begin{bmatrix} \mathbf{R} & \mathbf{J}_l(\boldsymbol{\theta}) \boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix}$.

Left/right Jacobians. With $\text{Exp}(\xi + \delta\xi) = \text{Exp}(\xi) \text{Exp}(\mathbf{J}_R(\xi)\delta\xi) = \text{Exp}(\mathbf{J}_L(\xi)\delta\xi) \text{Exp}(\xi)$,

$$\mathbf{J}_R(\xi) = \begin{bmatrix} \mathbf{J}_R(\theta) & \mathbf{0} \\ \mathbf{Q}_R(\xi) & \mathbf{J}_R(\theta) \end{bmatrix}_{6 \times 6}, \quad \mathbf{J}_L(\xi) = \begin{bmatrix} \mathbf{J}_L(\theta) & \mathbf{0} \\ \mathbf{Q}_L(\xi) & \mathbf{J}_L(\theta) \end{bmatrix}_{6 \times 6}, \quad (127)$$

and to first order, with $\xi^\wedge = \begin{bmatrix} [\theta \times] & \mathbf{0} \\ [\rho \times] & [\theta \times] \end{bmatrix}_{6 \times 6}$,

$$\mathbf{J}_L(\xi) \approx \mathbf{I}_6 + \frac{1}{2}\xi^\wedge, \quad \mathbf{J}_R(\xi) \approx \mathbf{I}_6 - \frac{1}{2}\xi^\wedge. \quad (128)$$

Homework. Find $\mathbf{Q}_L$ (and $\mathbf{Q}_R$).

Error states: SE(3) vs SO(3) + $\mathbb{R}^3$. Perturb $\mathbf{T}$ on the right by the SE(3) tangent $\delta\xi = [\delta\theta^\top, \delta\rho^\top]^\top$. Expanding with $\mathbf{J}_l(\delta\theta) \approx \mathbf{I} + \frac{1}{2}[\delta\theta \times]$ and $\delta\mathbf{R} = \text{Exp}(\delta\theta)$,

$$\begin{aligned} \mathbf{T} \delta\mathbf{T} &= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \text{Exp}(\delta\theta) & \mathbf{J}_l(\delta\theta)\delta\rho \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & (\mathbf{I} + \frac{1}{2}[\delta\theta \times])\delta\rho \\ \mathbf{0} & 1 \end{bmatrix} \\ &\approx \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\rho \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{R}\delta\rho + \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}. \end{aligned} \quad (129)$$

If instead we treat the pose as $SO(3) + \mathbb{R}^3$ (rotation and translation perturbed separately, $\mathbf{p} \rightarrow \mathbf{p} + \delta\mathbf{p}$),

$$\mathbf{T} = \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{p} + \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{p} + \mathbf{R}\mathbf{R}^\top\delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \mathbf{R}^\top\delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}. \quad (130)$$

Comparing the two conventions gives the relation between the error states (SE(3) tangent translation $\delta\rho$ vs. naive position error $\delta\mathbf{p}$),

$$\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}_{SO(3)+\mathbb{R}^3} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\rho \\ \mathbf{0} & 1 \end{bmatrix}_{SE(3)} \Rightarrow \begin{cases} \delta\rho = \mathbf{R}^\top\delta\mathbf{p} \\ \delta\mathbf{R}_{SE(3)} = \delta\mathbf{R} \end{cases}. \quad (131)$$

Adjoint. The adjoint transfers algebra elements across the group:

$$SO(3) : \mathbf{R} \text{Exp}(\varphi) \mathbf{R}^\top = \text{Exp}(\mathbf{R}\varphi) \Leftrightarrow \text{Ad}_{\mathbf{R}} = \mathbf{R}, \quad SE(3) : \mathbf{T} \text{Exp}(\xi) \mathbf{T}^{-1} = \text{Exp}(\text{Ad}_{\mathbf{T}} \xi), \quad (132)$$

with $\text{Ad}_{\mathbf{T}} = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ [\mathbf{p} \times] \mathbf{R} & \mathbf{R} \end{bmatrix}_{6 \times 6}$. The adjoint is itself a Lie group, with $\text{Ad}_{\mathbf{T}} = \exp(\xi^\wedge)$ and $\log(\text{Ad}_{\mathbf{T}})^\vee = \xi$.

Homework. Prove $\mathbf{T} \text{Exp}(\xi) \mathbf{T}^{-1} = \text{Exp}(\text{Ad}_{\mathbf{T}} \xi)$.

2.7.1 More operations

(1) Interpolation. For ${}^G\mathbf{T}_{t_1}, {}^G\mathbf{T}_{t_2}$, $t \in [t_1, t_2]$, $\lambda = \frac{t-t_1}{t_2-t_1}$: in $SE(3)$, with ${}^{t_2}\xi = \text{Log}({}^G\mathbf{T}_{t_1}^{-1} {}^G\mathbf{T}_{t_2})$,

$${}^G\mathbf{T} = {}^G\mathbf{T}_{t_1} \text{Exp}(\lambda {}^{t_2}\xi); \quad (133)$$

in $SO(3) + \mathbb{R}^3$, the rotation interpolates on the manifold and the position linearly:

$${}^G_t\mathbf{R} = {}^G_{t_1}\mathbf{R} \text{Exp}(\lambda \text{Log}({}^G_{t_1}\mathbf{R}^\top {}^G_{t_2}\mathbf{R})), \quad {}^G_t\mathbf{p} = {}^G_{t_1}\mathbf{p} + \lambda({}^G_{t_2}\mathbf{p} - {}^G_{t_1}\mathbf{p}). \quad (134)$$

(2) Time derivative. For $SE(3)$, with $\mathbf{T} = \mathbf{T} \text{Exp}([\delta\boldsymbol{\theta}; \delta\boldsymbol{\rho}]^\wedge)$ and body velocity, expand the limit using $\exp([\delta\boldsymbol{\theta} \times]) - \mathbf{I}_3 \approx [\delta\boldsymbol{\theta} \times]$ and $\mathbf{J}_l(\delta\boldsymbol{\theta}) \approx \mathbf{I} + \frac{1}{2} [\delta\boldsymbol{\theta} \times]$:

$$\begin{aligned} \frac{d\mathbf{T}}{dt} &= \lim_{\delta t \rightarrow 0} \frac{\mathbf{T}(t + \delta t) - \mathbf{T}(t)}{\delta t} = \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \left(\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \exp([\delta\boldsymbol{\theta} \times]) & \mathbf{J}_l(\delta\boldsymbol{\theta})\delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} - \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \right) \\ &= \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \exp([\delta\boldsymbol{\theta} \times]) - \mathbf{I}_3 & \mathbf{J}_l(\delta\boldsymbol{\theta})\delta\boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} = \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} [\delta\boldsymbol{\theta} \times] & \delta\boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} [\boldsymbol{\omega} \times] & \mathbf{v} \\ \mathbf{0} & 0 \end{bmatrix} = \begin{bmatrix} \mathbf{R} [\boldsymbol{\omega} \times] & \mathbf{R}\mathbf{v} \\ \mathbf{0} & 0 \end{bmatrix}, \end{aligned} \quad (135)$$

where $\mathbf{v}$ is the body velocity. For $SO(3) + \mathbb{R}^3$ the same computation with $\mathbf{T} = \begin{bmatrix} \mathbf{R} \exp([\delta\boldsymbol{\theta} \times]) & \mathbf{p} + \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}$ gives

$$\frac{d\mathbf{T}}{dt} = \begin{bmatrix} \mathbf{R} [\boldsymbol{\omega} \times] & {}^G\mathbf{v} \\ \mathbf{0} & 0 \end{bmatrix}, \quad (136)$$

with ${}^G\mathbf{v}$ the global velocity.

(3) Calibration. For IMU frame $\{I\}$ and camera frame $\{C\}$,

$${}^{C_1}_{C_2}\mathbf{T} = {}^C\mathbf{T} {}^{I_1}_{I_2}\mathbf{T} {}^C\mathbf{T}^{-1}, \quad \text{Exp}({}^{C_1}_{C_2}\boldsymbol{\xi}) = \text{Exp}(\text{Ad}_{({}^C\mathbf{T})} {}^{I_1}_{I_2}\boldsymbol{\xi}) \Rightarrow {}^{C_1}_{C_2}\boldsymbol{\xi} = \text{Ad}_{({}^C\mathbf{T})} {}^{I_1}_{I_2}\boldsymbol{\xi}. \quad (137)$$

Homework. Compute the Jacobians $\partial {}^{C_1}_{C_2}\boldsymbol{\xi} / \partial {}^{I_1}_{I_2}\boldsymbol{\xi}$ and $\partial {}^{C_1}_{C_2}\boldsymbol{\xi} / \partial {}^C\boldsymbol{\xi}$ (and the corresponding $SO(3) + \mathbb{R}^3$ versions).

2.8 $SE_2(3)$

Augmenting the pose with a velocity column gives the 5×5 group element

$$\Upsilon = \begin{bmatrix} \mathbf{R} & \mathbf{p} & \mathbf{v} \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix}, \quad \boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \\ \boldsymbol{\nu} \end{bmatrix} \rightarrow \boldsymbol{\xi}^\wedge = \begin{bmatrix} [\boldsymbol{\theta} \times] & \boldsymbol{\rho} & \boldsymbol{\nu} \\ \mathbf{0} & 0 & 0 \\ \mathbf{0} & 0 & 0 \end{bmatrix}_{5 \times 5}, \quad (138)$$

where (as in $SE(3)$) the tangent components $\boldsymbol{\rho}, \boldsymbol{\nu}$ are distinct from the group position/velocity $\mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho}$, $\mathbf{v} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\nu}$. The exponential, logarithm, and adjoint follow the same pattern as $SE(3)$:

$$\text{Exp}(\boldsymbol{\xi}) = \begin{bmatrix} \mathbf{R} & \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho} & \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\nu} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{p} & \mathbf{v} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad \text{Log}(\Upsilon)^\vee = \begin{bmatrix} \boldsymbol{\theta} \\ \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{p} \\ \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{v} \end{bmatrix}, \quad \text{Ad}_\Upsilon = \begin{bmatrix} \mathbf{R} & \mathbf{0} & \mathbf{0} \\ [\mathbf{p} \times] \mathbf{R} & \mathbf{R} & \mathbf{0} \\ [\mathbf{v} \times] \mathbf{R} & \mathbf{0} & \mathbf{R} \end{bmatrix}. \quad (139)$$

2.9 Quaternions

Quaternions are presented last because $SO(3)$, $SE(3)$, and the Lie-group view already cover most needs. There are two main conventions — *Hamilton* (right-handed) and *JPL* (left-handed); we use the **Hamilton** convention. For a thorough treatment see Solà [5] (Hamilton) and Trawny & Roumeliotis [4] (JPL).

A quaternion and its imaginary units satisfy

$$Q = q_w + q_x i + q_y j + q_z k, \quad i^2 = j^2 = k^2 = ijk = -1, \quad ij = k, \quad jk = i, \quad ki = j, \quad (140)$$

written as the 4×1 vector $\bar{q} = [q_w, q_x, q_y, q_z]^\top = [q_w, \mathbf{q}_v^\top]^\top$.

Product. The quaternion product is bilinear and can be written with left/right matrices:

$$\bar{p} \otimes \bar{q} = \mathcal{L}(\bar{p}) \bar{q} = \mathcal{R}(\bar{q}) \bar{p}, \quad \mathcal{L}(\bar{p}) = p_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{p}_v^\top \\ \mathbf{p}_v & [\mathbf{p}_v \times] \end{bmatrix}, \quad \mathcal{R}(\bar{q}) = q_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{q}_v^\top \\ \mathbf{q}_v & -[\mathbf{q}_v \times] \end{bmatrix}. \quad (141)$$

Identity, unit quaternion, inverse. The identity is $\bar{q} = [1, \mathbf{0}_{3 \times 1}^\top]^\top$. A rotation by angle θ about unit axis $\mathbf{k}$ is the unit quaternion

$$\bar{q} = \begin{bmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \mathbf{k} \end{bmatrix}, \quad \bar{q}^{-1} = \begin{bmatrix} \cos \frac{\theta}{2} \\ -\sin \frac{\theta}{2} \mathbf{k} \end{bmatrix}, \quad \mathbf{R}_{12} = \mathbf{R}_1 \mathbf{R}_2 \leftrightarrow \bar{q}_{12} = \bar{q}_1 \otimes \bar{q}_2. \quad (142)$$

Why $\theta/2$? — rotating a vector. With global frame $\{G\}$ and IMU frame $\{I\}$, rotate ${}^I \mathbf{p}$ via the sandwich product and convert to matrices with $\bar{p} \otimes \bar{q} = \mathcal{R}(\bar{q}) \bar{p}$ and $\bar{p} \otimes \bar{q} = \mathcal{L}(\bar{p}) \bar{q}$:

$$\begin{bmatrix} 0 \\ {}^G \mathbf{p} \end{bmatrix} = {}^G \bar{q} \otimes \begin{bmatrix} 0 \\ {}^I \mathbf{p} \end{bmatrix} \otimes {}^G \bar{q}^{-1} = \mathcal{R}({}^G \bar{q}^{-1}) \mathcal{L}({}^G \bar{q}) \begin{bmatrix} 0 \\ {}^I \mathbf{p} \end{bmatrix}. \quad (143)$$

Writing $\bar{q} = [q_w, \mathbf{q}_v^\top]^\top$ and multiplying the two matrices,

$$\begin{aligned} \mathcal{R}(\bar{q}^{-1}) \mathcal{L}(\bar{q}) &= \left(q_w \mathbf{I}_4 + \begin{bmatrix} 0 & \mathbf{q}_v^\top \\ -\mathbf{q}_v & [\mathbf{q}_v \times] \end{bmatrix} \right) \left(q_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{q}_v^\top \\ \mathbf{q}_v & -[\mathbf{q}_v \times] \end{bmatrix} \right) \\ &= q_w^2 \mathbf{I}_4 + 2q_w \begin{bmatrix} 0 & \mathbf{0} \\ \mathbf{0} & [\mathbf{q}_v \times] \end{bmatrix} + \begin{bmatrix} \mathbf{q}_v^\top \mathbf{q}_v & \mathbf{0} \\ \mathbf{0} & -\mathbf{q}_v \mathbf{q}_v^\top + [\mathbf{q}_v \times]^2 \end{bmatrix}, \end{aligned} \quad (144)$$

whose lower-right 3×3 block (acting on ${}^I \mathbf{p}$) gives

$${}^G \mathbf{p} = (q_w^2 \mathbf{I}_3 + 2q_w [\mathbf{q}_v \times] + \mathbf{q}_v \mathbf{q}_v^\top + [\mathbf{q}_v \times]^2) {}^I \mathbf{p}. \quad (145)$$

Using $[\mathbf{q}_v \times]^2 = \mathbf{q}_v \mathbf{q}_v^\top - (\mathbf{q}_v^\top \mathbf{q}_v) \mathbf{I}_3$ and the unit-norm condition $q_w^2 + \mathbf{q}_v^\top \mathbf{q}_v = 1$,

$$q_w^2 \mathbf{I}_3 + \mathbf{q}_v \mathbf{q}_v^\top + [\mathbf{q}_v \times]^2 = q_w^2 \mathbf{I}_3 + (\mathbf{q}_v^\top \mathbf{q}_v) \mathbf{I}_3 + 2[\mathbf{q}_v \times]^2 = \mathbf{I}_3 + 2[\mathbf{q}_v \times]^2, \quad (146)$$

so

$${}^G \mathbf{p} = (\mathbf{I}_3 + 2q_w [\mathbf{q}_v \times] + 2[\mathbf{q}_v \times]^2) {}^I \mathbf{p}. \quad (147)$$

Finally substitute $q_w = \cos \frac{\theta}{2}$, $\mathbf{q}_v = \sin \frac{\theta}{2} \mathbf{k}$, so $2q_w [\mathbf{q}_v \times] = 2 \cos \frac{\theta}{2} \sin \frac{\theta}{2} [\mathbf{k} \times] = \sin \theta [\mathbf{k} \times]$ and $2[\mathbf{q}_v \times]^2 = 2 \sin^2 \frac{\theta}{2} [\mathbf{k} \times]^2 = (1 - \cos \theta) [\mathbf{k} \times]^2$:

$${}^G \mathbf{p} = (\mathbf{I}_3 + \sin \theta [\mathbf{k} \times] + (1 - \cos \theta) [\mathbf{k} \times]^2) {}^I \mathbf{p} \quad (148)$$

$$= {}^G \mathbf{R} {}^I \mathbf{p} = \text{Exp}(\theta \mathbf{k}) {}^I \mathbf{p}, \quad (149)$$

recovering Rodrigues' formula — which is why the half-angle $\theta/2$ appears.

Derivative. With $\delta \theta = \delta \theta \mathbf{k}$ and small-angle $\delta \bar{q} = [\cos \frac{\delta \theta}{2}, \sin \frac{\delta \theta}{2} \mathbf{k}^\top]^\top \approx [1, \frac{1}{2} \delta \theta^\top]^\top$,

$$\begin{aligned} {}^G \dot{\bar{q}} &= \lim_{\Delta t \rightarrow 0} \frac{{}^G \bar{q}_{t+\Delta t} - {}^G \bar{q}_t}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{{}^G \bar{q} \otimes \begin{bmatrix} \cos \frac{\delta \theta}{2} \\ \sin \frac{\delta \theta}{2} \mathbf{k} \end{bmatrix} - {}^G \bar{q} \otimes \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{{}^G \bar{q} \otimes \begin{bmatrix} 0 \\ \frac{1}{2} \delta \theta \end{bmatrix}}{\Delta t} \\ &= {}^G \bar{q} \otimes \begin{bmatrix} 0 \\ \frac{1}{2} \boldsymbol{\omega} \end{bmatrix} = \frac{1}{2} \mathcal{R} \left(\begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix} \right) {}^G \bar{q} = \frac{1}{2} \begin{bmatrix} 0 & -\boldsymbol{\omega}^\top \\ \boldsymbol{\omega} & -[\boldsymbol{\omega} \times] \end{bmatrix} {}^G \bar{q} = \frac{1}{2} \Omega(\boldsymbol{\omega}) {}^G \bar{q}, \end{aligned} \quad (150)$$

where $\Omega(\boldsymbol{\omega}) \triangleq \begin{bmatrix} 0 & -\boldsymbol{\omega}^\top \\ \boldsymbol{\omega} & -[\boldsymbol{\omega} \times] \end{bmatrix}$.

Integration. Given ${}^G_t\bar{q}$, ω , δt , with $\delta\theta = \omega \delta t = \delta\theta \mathbf{k}$ and $\delta\bar{q} = [\cos \frac{\delta\theta}{2}, \sin \frac{\delta\theta}{2} \mathbf{k}^\top]^\top$,

$${}^G_{t+\delta t}\bar{q} = {}^G_t\bar{q} \otimes \delta\bar{q}. \quad (151)$$

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